LEARNING TO DISCOVER VALUE: VALUE-BASED PRICING AND SELLING CAPABILITIES FOR SERVICES AND SOLUTIONS

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Learning to Discover Value: Value-Based Pricing and Selling Capabilities for Services and Solutions

Abstract

Many manufacturers invest in advanced services and solutions to achieve superior customer value; however, research has only begun to examine capabilities for value-based pricing (VBP) and value-based selling (VBS) in relation to such offerings. This article explores (1) which capabilities firms seek to develop for VBP and VBS of industrial services and solutions and (2) how learning influences the development of these capabilities. An in-depth exploratory study of two global market leaders in their respective industries includes interviews with 66 respondents from the firms, as well as 12 interviews with customer and supplier informants, which reveal important capabilities for VBP and VBS. Higher-level learning supports value discovery, through dialogue with customers over time; this value in turn forms the basis for VBP and VBS. Higher-level learning capabilities also facilitate the adaptation and replication of the developed pricing and selling capabilities in various contexts.

Keywords: value-based pricing, value-based selling, value discovery, learning, industrial services, integrated solutions

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1. Introduction

As growing numbers of companies focus on providing comprehensive industrial services and customized solutions (Raja, Bourne, Goffin, Çakkol, & Martinez, 2013; Tuli, Kohli, & Bharadwaj, 2007; Ulaga & Kohli, 2018), they also acknowledge their growing need to foster capabilities to support value-based pricing (VBP) and value-based selling (VBS) efforts. In particular, VBP implies that the seller's pricing decisions account for customers' perceptions of the offering's use value, in relation to its price (Ingenbleek, 2014; Kienzler, 2018a). Capturing value from advanced offerings requires a greater focus on pricing (Parida, Sjödin, Wincent, & Kohtamäki, 2014). Then VBS refers to "a sales approach that builds on identification, quantification, communication, and verification of customer value" (Töytäri & Rajala, 2015, p. 101). Practitioner publications describe various pricing approaches and strategies to achieve better earnings (e.g., Chan, Jubas, Kordes & Sueling, 2015; Smith, 2016), reflecting managers' sense that they need to develop such capabilities. Existing research in turn cites their importance but also the difficulties that industrial companies have in developing (Hallberg, 2017a; Hinterhuber, 2008, 2017; Kowalkowski & Ulaga, 2017; Töytäri, Keränen, & Rajala, 2017) and learning, even when they have experience with service and solution agreements (Kowalkowski, Windahl, Kindström, & Gebauer, 2015; Storbacka, 2011).

Most prior literature separates VBP (Dost & Geiger, 2017; Kienzler, 2018b; Liozu & Hinterhuber, 2014) from VBS (Terho, Haas, Eggert, & Ulaga, 2012; Terho, Eggert, Haas, & Ulaga, 2015; Töytäri, Alejandro, Parvinen, Ollila, & Rosendahl, 2011) and assumes they are static concepts. However, Töytäri et al. (2015) explore their interconnections, with the argument that successful VBP requires proactive efforts to influence customer value perceptions, through sales processes. Moreover, VBS can only capture sufficient created value if the corresponding revenue mechanisms or profit formulae are effective (Johnson,

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Christensen, & Kagermann, 2008; Sawhney, 2006; Töytäri et al., 2015). This interdependency represents fertile ground for further research, particularly to clarify which pricing and selling capabilities are necessary for industrial services and solutions to generate profits. In parallel, insufficient attention has centered on how firms can modify their routines to match different operating environments, using dynamic (higher-level) capabilities (Teece, Pisano, & Shuen, 1997; Zollo & Winter, 2002). Learning thus is an important consideration (Easterby-Smith & Prieto, 2008), especially in relation to the scope of adaptation and replication (Winter & Szulanski, 2001).

Therefore, the current study explores VBP and VBS capabilities in an industrial service and solution context. Through an extensive study of two global market leaders, we identify key VBP and VBS capability dimensions, as well as how learning influences them to support reconfigurations and replications in different contexts. Accordingly, in addition to specifying the dimensions of pricing and selling capabilities and the process by which they develop, this study complements existing literature by explaining the critical role of learning, such that providers engage in relational processes and actively include customers in value discovery through dialogue over an extended period.

2. Theoretical Background

2.1 Capabilities perspectives and learning

In line with the resource-based view of a firm (Wernerfelt, 1984), organizations require capabilities to conduct their activities (Richardson, 1972); an "organizational capability refers to the ability of an organization to perform a coordinated set of tasks, utilizing organizational resources, for the purpose of achieving a particular result" (Helfat & Peteraf, 2003, p. 999). Organizations develop capabilities by assembling available resources into specific, unique configurations that enable them to transform input into more valuable output (Amit &

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Schoemaker, 1993). The imperfectly imitable, firm-specific resources provide organizations with competitive advantages (Barney, 1991); the resulting capabilities take different forms.

For example, an operational capability "enables a firm to perform an activity on an on-going basis using more or less the same techniques on the same scale to support existing products and services for the same customer population," (Helfat & Winter, 2011, p. 1244) and such strategic replications of routinized activities can be a source of competitive advantage (Winter & Szulanski, 2001). Higher-level (or dynamic) capabilities instead "enable a firm to alter how it currently makes its living" (Helfat & Winter, 2011, p. 1244). According to Helfat and Winter (2011), capabilities supporting existing business processes, such as abilities to replicate or proliferate, have dynamic attributes, such that they occur through complex, interwoven networks of routines and resources. For example, template replication requires dynamic capabilities at headquarters, but the development of the template likely occurs through learning that informs the logic of replication (Winter & Szulanski, 2001). Such organizational learning processes enable firms to identify opportunities, develop offerings, and reconfigure to establish new modes of value creation, delivery, and capture. Because capabilities thus appear rooted in organizational learning (Easterby-Smith & Prieto, 2008), a dynamic capability reflects "a learned and stable pattern of collective activity through which the organization systematically generates and modifies its operating routines in the pursuit of improved effectiveness" (Zollo & Winter, 2002, p. 340). That is, learning leads to capability development, which then can provide sustained competitive advantages.

In the learning literature, such "higher-level" learning supports searches for new ways of doing things (Cope, 2003). It also might entail a specific process that allows for experimentation, repetition, and the identification of new opportunities (Bowman & Ambrosini, 2003; Teece et al., 1997), which can involve "learning from peers; learning by doing; learning from feedback from customers and suppliers; learning by copying; learning

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by experiment; learning by problem solving and opportunity taking; and learning from making mistakes" (Gibb, 1997, p. 19). Engeström (2014) proposes expansive learning as another form of higher-level learning based on the activity theory framework. This process of ongoing learning actions entails questioning the present activity, analyzing a given situation and needs for change, modeling the new activity, applying the new activity, and consolidating and reflecting on the new activity (Engeström, 2014; Engeström & Kerosuo, 2007).

2.2 VBP and VBS capabilities for industrial services and solutions

Building on a capability view, scholars highlight the strategic significance of pricing capabilities (Dutta, Bergen, Levy, Ritson, & Zbaracki, 2002; Johansson, Hallberg, Hinterhuber, Zbaracki, & Liozu, 2012; Liozu & Hinterhuber, 2013b). For example, price-setting capabilities constitute complex routines, skills, systems, coordination mechanisms, and complementary resources that exist both within the firm (identifying competitors' prices, setting pricing strategy, translation from pricing strategy to price) and in relation to customers (convincing customers of price change logics, negotiating price changes with major customers) (Dutta, Zbaracki, & Bergen, 2003). Pricing activities based on these configurations can lead to superior profitability and competitive advantages (Liozu, 2015a; Liozu & Hinterhuber, 2013a). Accordingly, Johansson et al. (2012) argue that capabilities are fundamental to successful pricing once the firm acknowledges pricing as an organizational effort. Unlike other resources, such capabilities cannot be bought but must be developed over time and embedded in the organization (Liozu, 2015a).

2.2.1 Service and solution selling. Research on industrial services details many of the selling capabilities required for such offerings (Ulaga & Loveland, 2014), as well as some of the needed pricing capabilities (Rapaccini, 2015). Literature on solution selling is also relevant here, because solutions are "longitudinal relational processes, during which a solution provider integrates goods, service and knowledge components into unique

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combinations that solve strategically important customer specific problems, and is compensated on the basis of the customer's value-in-use" (Storbacka, 2011, p. 699). It generally takes more time to sell advanced services and solutions (Tuli et al., 2007), reflecting the wider range of needs, processes, and uncertain outcomes (Ulaga & Kohli, 2018). Moreover, "the development of an effective pricing process (i.e., setting, changing and negotiating prices) is … time-consuming, costly and complex with significant organizational and informational barriers restricting the process" (Hallberg, 2017a, p. 180). Dialogues and interactions with customers are therefore crucial considerations (Kindström, Kowalkowski, & Alejandro, 2015; Tuli et al., 2007), and further research is needed (Terho et al., 2012).

Among the various industrial services and solutions studies of capability development (for overview see Eloranta & Turunen, 2015), Ulaga and Reinartz (2011) propose a framework of five capabilities essential to the successful deployment of industrial services (process service-related data, assess and mitigate service-specific risks, design offerings around service and not technology, sell services as opposed to products, and engage in effective service deployment), suggesting the importance of service sales capabilities. Steiner, Eggert, Ulaga, & Backhaus, (2016) also find that advanced marketing capabilities—such as understanding, documenting, and communicating customer value—are required for manufacturers to sell industrial services profitably. Storbacka (2011) identifies 12 categories of 64 capabilities necessary for solutions, some of which pertain to selling solutions. Both Kindström et al. (2015) and Ulaga and Loveland (2014) cite critical demands: to gain deep understanding of a customer's business model, to coordinate multiple needs and competencies in vendor and customer organizations, and to proactively manage expectations and visualize value over time. Ulaga and Loveland (2014) also emphasize the importance of VBS to help customers understand the value of the (intangible) service elements. Although service selling and VBS are often linked to the sales of complex offerings, such that vendors

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get compensated on the basis of the customer's value-in-use (Crosby, Evans, & Cowles, 1990; Storbacka, 2011), Kindström et al. (2015) argue that VBS actually requires a service-centric rather than a product-centric mindset in the first place. The applicability of VBS also may depend on customer types and stages in the buyer–seller relationship (Terho et al., 2012), yet traditional after-sales activities tend to adopt a cost-plus–based pricing approach (Kowalkowski & Ulaga, 2017; Töytäri et al., 2015). Finally, VBS is geared toward the sales force level, but service selling also likely considers the sales function level (Kindström et al., 2015).

2.2.2 Value-based pricing and selling. Prior research identifies specific stages for VBS. Extending two-stage capability frameworks (i.e., planning and implementation; Eades, 2004; Storbacka, 2011; Terho et al., 2012), Töytäri and Rajala (2015) propose three adjacent VBS stages: planning, implementation, and leverage, each of which requires salient capabilities and managerial practices, such as segmentation and stakeholder identification, value analysis, value proposition development, sales tools preparation, value communication, value quantification, and value verification.

Similar to VBP, which involves the extent to which a customer perceives the benefits from an offering with respect to the price (Ingenbleek, 2014; Kienzler, 2018a), a value-based selling logic requires focusing less on product and service attributes and more on understanding what customers value (Anderson & Narus, 1998), which may be manifest in their business models and processes (Frandsen, Boa, & Raja, 2019; Töytäri et al., 2017;). Value in this sense is the difference between the buyers' willingness to pay and the opportunity costs of suppliers (Brandenburger & Stuart, 1996). In terms of profit potential, VBP is superior to cost-based approaches that set price based on costs to generate a certain return on investment (Liozu, 2015b), as well as to competition-based approaches that depend on competitors' pricing or positioning (Liozu, 2015b; Liozu & Hinterhuber, 2013a; Morris &

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Fuller, 1989; Nagle, Hogan, & Zale, 2016). Such pricing capabilities also enhance the collective confidence of the sales force (Liozu, 2015a).

As a sales approach that builds on the identification and quantification of customer value-in-use (Anderson, Kumar, & Narus, 2007; Töytäri and Rajala, 2015), VBS is closely linked to VBP. It aims to communicate how the offered service is valuable so that the customer ultimately expresses greater willingness to pay, because the price model is well justified (Terho et al., 2012). In this sense, VBS enables VBP; it reflects the firm's ability to capture the value promised to the customer (Bowman & Ambrosini, 2000; Töytäri & Rajala, 2015; Töytäri et al., 2015). Accordingly, the two concepts are "inextricably linked," and "one does not exist without the other" (Moorman, 2013, p. 335). A related argument holds that a customer's business model must be input for both VBP and VBS (Terho et al., 2012; Töytäri & Rajala, 2015), especially in an industrial services context. Advanced services require both VBS and VBP (Kowalkowski & Ulaga, 2017), but it may be difficult for the seller to understand and quantify customers' value perceptions before entering the actual selling process (Kindström et al., 2015). It is thus surprising that, with some notable exceptions (see Liinamaa, Viljanen, Hurmerinta, Ivanova-Gongne, Luotola, & Gustafsson, 2016; Töytäri et al., 2015), most research presents VBS and VBP separately, without examining how learning influences them, despite the clear importance of learning activities for value capture (Parida et al., 2014).

Thus, higher-level learning requires further consideration, with regard to how organizations develop and implement VBS and VBP capabilities in a qualitatively new way to function. Ongoing learning actions shape the VBS and VBP capabilities and are worthy of further research, to move beyond literature that emphasizes pricing capabilities (Dutta et al., 2003; Hallberg, 2017a; Kindström et al., 2015; Liozu & Hinterhuber, 2013a, b) and address pricing *and* selling capabilities based on value. Building on these related streams of literature,

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we leverage a context of industrial services and solutions to explore how industrial firms develop pricing and selling capabilities and the influence of their learning processes on VBP and VBS. The central research questions are as follows:

1. What capabilities do firms seek to develop for VBP and VBS for their industrial service and solution offerings?
2. How does learning influence VBP and VBS for industrial services and solutions?

3. Research Methodology

Noting the complexity of adopting and implementing VBS and VPS approaches, an in-depth exploratory case-based research approach (Yin, 2009) is appropriate, with an abductive research process (Dubois & Gadde, 2002) featuring iterative shifts between extant literature and empirical observations. As part of a larger research project examining the competitiveness of multinational organizations with Danish origin, the authors negotiated close access to two firms—traditional manufacturing organizations transitioning toward service business models. During the study, the empirical findings prompted the closer focus on VBP and VBS for industrial services and solutions, due to the related challenges that the case organizations faced. Prior VBP and VBS literature informed the data collection, and then the gathered insights informed subsequent steps.

3.1 Sampling approach

We applied a purposeful sampling approach (Miles & Huberman, 1994; Patton, 2015). Of the many sampling strategies available (see Patton, 2015), we adopted emergent theory sampling, examined constructs, and elaborated on them (Miles & Huberman, 1994). The two case companies, Alpha and Beta, and respondents were selected on the basis of specific criteria. In particular, both case companies (1) were changing their business models to become more service- and solution-oriented, (2) had adopted a value-driven mindset for their offerings, (3) recognized the importance of VBP and VBS, (4) were actively developing capabilities and

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processes to support this implementation, and (5) provided access to the key informants, who in turn had to have knowledge of pricing, secondary data sources, and customer and supplier firms. The initial exploratory approach, applied to Alpha, identified emergent issues and associated literature, and then we purposefully incorporated the Beta case to sharpen the theory analysis process (Patton, 2015).

Miles and Huberman (1994) describe two important tasks of qualitative sampling. First, clear boundaries must be set for the cases. We needed a high level of access and openness from the case companies to explore VBP and VBS, which many firms are reluctant to grant, due to the sensitivity of the required information. Therefore, we built trust with the firms over time. Second, a frame should enable uncovering, confirmation, or qualification of the basic processes or constructs within a research study. In line with our approach, which constantly moves between data and theory in an abductive manner (Dubois & Gadde, 2002), we progressively combined emergent insights with theory to extend understanding of the focal phenomena (Strauss & Corbin, 1990).

3.2 Case overviews

Alpha and Beta are global market leaders in their respective industries. Alpha is a large engineering firm in the maritime industry, headquartered in Denmark. It has been in operation for more than 30 years. It recently expanded by locating offices in Singapore to support its operations in Asia. It provides high-tech installations to firms in marine, shipbuilding, and wind industries throughout the world. The study data came from a large division of the company that provides systems for onshore and offshore telecommunications and firefighting, responsible for its own profits and losses. The division has approximately 220 employees, and though its shift to services is fairly recent (previously, service support was provided by project engineers), it operates a fully functioning service department. Alpha

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successfully sells value-based service agreements for comprehensive services to implement integrated solutions in the offshore wind industry.

Beta is a large engineering firm that provides entire plants for the mining and cement industries and has been in operation for over 100 years. It owns many product companies that provide capital-intensive equipment for those industries, and it has grown significantly in recent years, primarily through acquisitions. A leading brand, it has built a reputation for engineering excellence. The company's matrix structure contains four main divisions, each with profit and loss responsibilities. It depends on large projects, typical of companies in the mining and cement industries, and it is susceptible to the cyclical demand for projects based on commodity prices. After the global financial crisis, the mining and cement industries experienced decreased demand, which has limited the number of large projects in most markets. As a result, services and solutions have received greater attention, as potential cushions of the effects of decreased demand for large projects by providing through-life value to customers. The firm's service portfolio ranges from provision and management of spare parts to consulting and training customers' employees. In addition, Beta provides extended operation and maintenance solutions, sometimes for the lifetime of a plant. Today, the company successfully sells comprehensive service agreements for services tied to specific equipment or full operation and maintenance agreements for entire plants.

Table 1 provides an overview of both focal firms. We also collected data from key supplier and service customer firms of Alpha, including providers of firefighting systems (Suppliers 1 and 2) and telecommunication equipment (Supplier 3) and customers such as energy utility companies (Customers 1 and 2) and a drilling operator (Customer 3). For Beta, we interviewed a customer in the mining industry (Customer 4). These interviews provide important supplementary insights.

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3.3 Data collection

At Alpha, the first round of interviews was conducted on-site during May 2015. These exploratory interviews aimed to shed light on Alpha's shift toward service provision and the associated challenges (see Appendix A for the interview guide). We started with a group interview with key informants to gain an overview, during which pricing and selling issues emerged as highly significant. Following the initial data collection round, we delved into relevant literature, with a particular focus on industrial services and solutions. With a subsequent data collection in December 2015, we systematically addressed pricing issues but again encountered some unexpected and relevant observations. Additional data collection involved respondents in Singapore and Denmark in 2018. The data collection continued until we achieved an in-depth understanding of the issues (Denzin & Lincoln, 2000). Most of these efforts involved semi-structured interviews, such that in total, we conducted 26 interviews within one division of Alpha. In addition, we had access to documentation and presentations, took field notes based on informal communications, and held regular feedback sessions (both informal and as workshops) with management and staff responsible for service development.

At Beta, the data collection has been ongoing since December 2015. We built on the interview template developed during data collection at Alpha. As with Alpha, we began with a group interview with staff responsible for the service part of the business to gain an overview of the organization. As new insights emerged during data collection, we consulted extant literature and explored issues with respondents. This stage was abductive; there was constant movement between empirical data and theory. The 40 semi-structured interviews involved employees from the different regions in which the company operates. Table 2 lists the respondents and interview durations. As with Alpha, both informal and formal feedback sessions helped validate the findings. This important step encouraged clarification and elaboration on the issues with key informants.

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In addition, a number of "adjunct" interviews (Piekkari, Welch, & Paavilainen, 2009) involved three supplier and three customer organizations of Alpha and one customer organization of Beta. These interviews created a more holistic view, rather than relying solely on the focal firms' perspective, which expands our understanding of value. We conducted 12 interviews with respondents from these seven firms, as detailed in Table 2.

INSERT TABLE 2 ABOUT HERE

3.4 Data analysis

With the abductive approach, we performed data analysis throughout the study. All the data were organized and coded by multiple researchers using NVivo software. First, the research team conducted an in-case analysis of Alpha, then coded and analyzed the Beta data. The initial coding template reflected extant VBP and VBS and capabilities literature (e.g., Hinterhuber, 2008; Terho et al., 2012; Toytari & Rajala, 2015). As the coding and data analysis progressed, we inserted, deleted, and redefined various codes (King, 2004; Strauss & Corbin, 1990). Two researchers, involved in the data collection, and a third independent researcher coded the data independently (Miles & Huberman, 1994). The three coders extensively discussed the codes and coding structure, then brought any disagreements to the other researchers involved in the project until we attained agreement regarding the codes and labels. The research team engaged in constant dialogue, debating similarities and differences across the empirical findings and extant literature. The final data structure is in Fig. 1. Although we present the findings in a structured, linear way to facilitate understanding, the analysis process was nonlinear, involving simultaneous collections, coding, and analyses of the data (Strauss & Corbin, 1990). We constantly moved among emerging insights and extant literature while analyzing and reanalyzing these data.

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3.5 Reliability, validity, and trustworthiness

Reliability, trustworthiness, and validity are common concerns for qualitative research, so we assessed the collected data and resulting analysis results according to established criteria for qualitative research (see Table 3).

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4. Findings

Both firms regard industrial services and solutions as significant opportunities for growth, though Beta has moved farther along toward providing integrated offerings, likely due to its longer history. Economic turbulence and heightened competition have caused both firms to suffer lack of demand from traditional customer bases; they regard advanced services as alternative means to generate stable revenue, with less vulnerability to the peaks and troughs of their project-based industries. This section presents detailed descriptions of the VBP and VBS capability dimensions (Appendix 2 contains relevant quotes). In line with an abductive logic, we discuss four aggregate dimensions that emerge from the data in relation to evidence from extant literature: customer identification and analysis, implementation of VBP and VBS, verification of value phases, and the influence of learning on the phases.

4.1 Customer identification and analysis

The VBP and VBS approaches are potential means for firms to move beyond simply discussing prices with customers to address value. In an initial phase, this shift necessitates the development of both VBP and VBS dimensions (customer segmentation; customer value analysis) and VBS-specific sales tools and value propositions.

4.1.1 Customer segmentation. Anderson and Narus (1998) and Töytäri and Rajala (2015) highlight the common practice of using a customer process analysis to identify influential stakeholder groups within a target firm, to create segment-specific value propositions. Determining which stakeholder groups to target was challenging for both Alpha

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and Beta. To varying extents, they had routines for tracking pricing and an understanding of customers' price sensitivity. Beta systematically segmented its customer base and was better able to identify which customers might be interested in VBP and VBS, yet customer segmentation still proved challenging, due to differences in customers' ownership structures and maturity levels, across industries and regions. Alpha faced different challenges, such as a lack of knowledge about end customers and restrictions on engaging in dialogue with them.

The extent to which delivered services added value for customers also varied significantly. Both firms recognized that technological competencies and attitudes toward maintenance and service activities were key to segmenting customers; the customer must understand why advanced industrial services might be beneficial and then be open to allowing the provider to deliver those services. Beta developed a conceptual model that linked different customer segments to specific service offerings in its attempt to strategically price and sell services based on value. Thus, in addition to revenue, the firms recognized customers' own competencies and propensity to value advanced industrial services, as elemental to their segmentation efforts. Coherent service portfolios also should reflect a solid foundation of customer insights, so that any subsequent reconfigurations address areas of significant value.

4.1.2 Customer value analysis. Firms engaging in VBP and VBS processes commonly seek to gain thorough insights into their customers' production and value-creation processes to identify opportunities for improvement (Töytäri & Rajala, 2015). Beta had routines in place to conduct a detailed value analysis of its customers' processes, such as mapping flow processes for their plants. For each customer, it gained the ability to deduce which processes could be more efficient, how much value could be added, and which solutions to present. In support of these analyses, Beta developed specific routines and templates for conducting and reporting plant audits, commercialized as a service offering.

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Moreover, Beta trained service personnel to conduct routine audits and transfer relevant information to sales to identify business opportunities. For its value analyses, Alpha required access to customer equipment data, which were not always available, so it could not conduct thorough analyses of all its customers.

Despite being time-consuming and challenging, collecting and analyzing data from customers to develop insights is essential to pricing and selling processes. The findings suggest that a customer value analysis should precede efforts to convince customers of the price change logic or negotiate price changes, which Dutta et al. (2003) cite as two steps of a pricing capability. Both Alpha and Beta recognize that high-level access to customers is vital for customer value analyses and have routines in place for maintaining customer purchase histories, analyzing the value being delivered, and considering the potential for doing so.

4.1.3 Development of sales tools. Both Alpha and Beta recognize the importance of developing necessary tools to communicate the value of service and solution offerings. In some parts of its business, Beta maintains sales support instruments, such as marketing materials (e.g., brochures explaining full flow sheet capabilities), technology-enabled devices, plans for service innovations, and tools for calculating the total cost of ownership (TCO) for projects over their lifetime. Some tools solely supported sales, though their use differed across regions. For example, one of Beta's European offices produced a computer-animated movie to demonstrate the requirements for tendering for a complicated service contract; at its Danish headquarters, 3D-printed models of equipment helped explain what different services entailed. In other regions, sales departments integrated sensors and other technologies to predict part failure rates and use the accrued knowledge to sell new parts and plant overhaul services. However, no sales tools were leveraged globally, and some were considered irrelevant for certain regions or customers. For example, 3D printing requires equipment and specialized knowledge investments, which were not available at all locations.

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Both Alpha and Beta needed to develop new sales tools for services, beyond those they might have used for traditional sales. To communicate what the offerings entailed and define value for customers, they developed processes to standardize service concepts, which are important tools for VBP and VBS. In addition, at Alpha, VBS was complicated by the absence of sufficiently standardized service concepts; the urgent need to develop them was widely recognized. To address this need and communicate about services internally and externally, Alpha developed a global service portfolio, based on customers' identified needs, supported by a cross-departmental development team that established routines for developing service offerings and related sales tools. At Beta, the central service organization created routines for identifying service offerings developed throughout the global organization, then sought to align those routines with the global service portfolio to support the development of sales tools. In both cases, tools enable the VBS processes (Töytäri and Rajala, 2015), such as IT-based systems that constitute commercial decision resources (Hallberg, 2017b).

4.1.4 Value proposition development. Previous studies identify the need to develop value propositions carefully for VBS (Anderson et al., 2006; Terho et al., 2012; Töytäri and Rajala, 2015). Both Alpha and Beta recognize a key challenge: a lack of full integration in the value proposition provided to the customer. Unless specifically required in the tender, services rarely were included in the bid or proposal stages, with clear implications for the firms' ability to increase their service business, because customers simply remained unaware of the value propositions and potential benefits. Selling services at a later stage was more difficult; thus, they needed to change organizational mindsets about introducing services as part of the value proposition earlier in the process.

Alpha developed routines to include service engineers earlier, such as in the tendering phase and factory acceptance test, when customer representatives typically are present and thus can be made aware of the benefits. Additional obstacles arose when projects involved

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intermediaries, such as engineering, procurement, and construction (EPC) contracts, or contained project clauses that withheld the identity of the end customer. Contractual prohibitions on interacting with end customers hindered introductions of value propositions for services and solutions, because it needed knowledge of the end customer to identify key benefits in relation to specific customer processes and contexts, as well as predict customers' willingness to pay based on value perceptions (Nagle & Müller, 2017).

4.2 Implementation of VBP and VBS

Both providers directed particular attention to the implementation phase, which required fostering trust and building reputation, as well as value creation and communication.

4.2.1 Fostering trust and building reputation. Customers' perceptions of the trustworthiness and reputation of the firm stem from confidence, and developing trust and reputation is especially important for firms using a VBP and VBS approach, as a means to gain access to critical information about customer processes and encourage perceptions of fairness. As Töytäri and Rajala (2015) predict, this dimension is key in the implementation phase, and trust and credibility can be developed through brand image and reputation. Both Alpha and Beta had reputations for delivering highly engineered equipment and systems, having fostered customers' trust in the brands over many years with relational approaches. In addition, they could demonstrate a history of exceptional delivery. Beyond general trust in a provider's capabilities, relationships with and the performance of individual service engineers offer sources of trust. In many cases, service engineers are privy to competitive price information related to competitors' offerings that they gain from customers and provide to the pricing team. However, in the event of a service engineer's error, trust can diminish.

4.2.2 Value creation and communication. Value creation refers to the perceived value-in-use for the customer. For Alpha and Beta, demonstrating value creation was essential to the implementation phase. Service and solutions offerings typically continue over

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time, so it is necessary to demonstrate value creation at regular intervals. Customers appeared quick to forget the benefits of what was being delivered, such as engineers' technical expertise, which provided those customers with cost savings by decreasing their required time, effort, and resources. Furthermore, customers needed to perceive the distribution of created value as fair; otherwise, it would have negative ramifications for their service contract renewal. The firms learned that merely creating value through resources was not sufficient; they had to communicate value as well. When they did so, they could expand and refine their abilities to price and to sell services and solutions based on value-in-use (Hinterhuber, 2008; Terho et al., 2012; Töytäri et al., 2015). Respondents further suggested that value communication needed to start early in the sales process, prior to the implementation phase, supported by effective internal communication about value (Töytäri et al., 2017).

Both Alpha and Beta recognized the need to identify the central decision maker in the customer's buying center; after identifying this person, they still struggled with access though. In addition, their communication had to be tailored to the contact's level of technical expertise, which was not possible if the company lacked sufficient knowledge. To begin a conversation with a customer about the value-based offer, Alpha and Beta had to identify where value could be created. Providing clear messages requires value-based sales and marketing material, which was not always available or developed yet. On an ad hoc basis, Beta created marketing material in some parts of its business to communicate value, but it did not have a complete overview. Beta also focused on developing metrics, related information systems, and routines for capturing performance enhancements to enable value creation, as well as to demonstrate and communicate such value to customers.

Both Alpha and Beta struggled with internal incongruences regarding how services should be priced and sold based on value. Insufficient internal communication among departments intensified the problem, so both firms established headquarter-based initiatives.

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Specialists and senior-level managers began to develop and communicate service offerings, along with a technological infrastructure to support advanced services. As part of the initiatives, they developed documentation routines for service delivery and refined measures of the outcomes of interventions.

4.3 Verification of value

The notion of value can be elusive. The case companies exhibited strong recognition of the importance of an ability to verify the value created and routines to reconfigure in response to different contexts. Specifically, the verification of value involves quantification and value assessment, value capture, and knowledge management.

4.3.1 Quantification and value assessment. Ongoing analytical processes should evaluate and verify customer value (Anderson et al., 2006; Hinterhuber, 2004). At both Alpha and Beta, when reconfiguring pricing for different customers, it was necessary to quantify and make the benefits visible, which was challenging. Guarantees and references represent one tactic; documentation of services delivered also reminded customers of what was being provided, so that it facilitated resale processes. The TCO tools (Section 4.1.3) quantified the value of delivered services and solutions. Because value is neither static nor objective (Eggert et al., 2018) though, quantification also must reflect the fluid nature of customer perceptions of the value of the offering. These perceptions also need to inform the use of tools such as TCO, which specifies the relationships of specific offering characteristics with the costs incurred by customers. Changing equipment features or adding related services typically decreases these costs, which may provide the customer with enhanced value. To the extent that the customer accepts the calculation, TCO tools allow for the quantification of value.

Typically, value is assessed throughout the implementation but is especially important toward the end of a contract. Insufficient assessments of delivered services may lead to customer skepticism about the service contract's overall benefit. Alpha on occasion granted

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customers permission to use new technology on a trial basis to assess the value, which then provided a basis for discussions of price. Key performance indicators (KPIs) are also crucial, though the changing natures of the industries in which Alpha and Beta operate makes their assignment difficult. The KPIs often change on short notice, which also might mean that VBP and VBS are not be appropriate in all situations, such that profitability depends on the context. The customer's perception of value can adapt to changing business environments too. For example, Beta was asked to achieve specific KPIs for outcome-based contracts but could not do so, due to a drop in oil prices and customers' lowering or stopping production output.

Thus, in some cases, it may be better to sell services on a cost-plus basis rather than applying VBS. Respondents cautioned that selling services based on value or output can result in dramatic losses if the underlying contract is inappropriately designed (Liinamaa et al., 2016; Renault, Dalsace, & Ulaga, 2010). They must recognize what approach is most appropriate for each context with each customer, on an ongoing basis. This recognition in turn requires defining criteria for evaluating performance. At Beta, it sought to establish routines to learn from negotiation processes and prior contracts, so that it could mitigate risks from unforeseen changes in market conditions and customer operations that relate to contract criteria. For contracts tied to output, such routines might include addressing in advance how metrics can be renegotiated if aggregate demand changes, which then affects the customers' production volumes and planning preferences.

4.3.2 Value capture. Processes can allow firms to claim some of the value created in the exchange relationship (Kienzler, 2018b). For providers, a crucial aspect of VBP is capturing a fair share of the value created for the customer (Töytäri et al., 2015), though shifting from providing certain services for free to charging for them is often difficult (Witell & Löfgren, 2013). To create and ensure sufficient profits, the company must enforce its

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prices. At Alpha, representatives found it difficult to differentiate the value of service contracts from the value delivered by noncontractual services. The former does not necessarily enhance customer-perceived value if the customer can get the same service for less money and without a binding agreement, which decreases its willingness to pay. Sales staff at Alpha often would promise services free of charge to important customers with which they had long-term relationships. Although such free service provision strengthened the bond between Alpha and its customers, it made it more difficult for the company to establish the value of its service offerings, distinct from the value of free-of-charge services. Profit decreased due to missed opportunities by sales staff, and Alpha often found it difficult to convince customers to engage in service contracts, because they did not perceive any added value when an equal standard of service historically had been provided for free. Both Alpha and Beta realized though that the value of more comprehensive service agreements resonated with customers' needs for certain outcomes over extended periods. With service agreements, Alpha could capture value by charging for periodic maintenance activities, and Beta promised comprehensive operation and maintenance agreements.

4.3.3 Knowledge management. Knowledge management activities permit a firm to manage and share pricing and selling know-how. Each company maintains computer software tools that engineers can use on-site to support knowledge sharing efforts and capture customer needs. However, the most valuable channels for sharing customer insights and best practices remained informal interactions among employees in situated settings. Still, several factors hindered knowledge sharing, including the silo mentality that existed in the project, product, and service divisions of both Alpha and Beta. By barring effective knowledge sharing, these silos also limited the firms' ability to price and sell services. In addition, vast differences characterized the various businesses divisions, due to their unique histories and growth acquisitions, as well as language differences. A lack of familiarity across divisions

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thus impaired VBP and VBS in both firms. For example, engineers from one division could not recognize VBP and VBS opportunities for other divisions, because they did not know which types of services were offered. Both firms sought to address these issues with initiatives to identify opportunities for adapting services developed with specific customers to other regions and divisions. Headquarters of both firms identified local service innovations and included them in global service programs. Moreover, the documentation of equipment configurations was essential to both Alpha's and Beta's product offerings, and they both had extended systems, such as online monitoring of plant performance or diagnostic software installed in equipment, that could capture information related to clients' uses of equipment. The systems and associated analytical routines create potential to offer enhanced value to customers and support efficient internal coordinations of service delivery.

Table 4 summarizes how the themes relate to VBP and VBS across the three phases. Although prior literature distinguishes VBP from VBS, we find in practice that the two are closely interrelated, with continuous interplays among their relevant capabilities. The process of developing and deploying capabilities is not linear but rather iterative. Overall then, different VBP and VBS capabilities relate to a general need to embed learning processes, underpinned by value discovery and relational approaches, as we elaborate next.

INSERT TABLE 4 ABOUT HERE

4.4 Learning

Learning processes in both firms relate strongly to capabilities for VBP and VBS across the customer identification and analysis, implementation, and verification phases. Such learning through VBP and VBS has business-level implications, because the knowledge thus acquired influences current and future service offerings, as well as value for customers. Although their knowledge enabled the two providers of industrial equipment to develop advanced services, they needed VBP- and VBS-instigated processes with customers to discover value. That is,

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VBP and VBS pertain to setting and adjusting prices to reflect the value created, but they also constitute a critical method for providers to engage in experimentation and dialogue with customers regarding value, whether potential and realized.

4.4.1 Value discovery. Pricing industrial services based on value requires some mutual understanding of value, whether available or possible to negotiate. The provider, due to its knowledge of the impact of both equipment and services, is in a good position to develop value-creating offerings, but realizing value depends on usage and service execution within the customer sphere. A range of variables and relationships can affect the realization of value; learning related to these variables and relationships can reveal unrecognized potential value that might be realized through additional services, beyond service agreements and contracts. Engaging in value discovery with the customer also places the provider in a stronger position, because it gains superior knowledge of value constellations (i.e., different elements and actors involved), which might create entry barriers for competitors. On the customer side, value discovery provides experience with the importance of service agreements, which then can reveal missed or potential opportunities for pricing and selling based on value. Beta discovered, through its relational approach, that the service it provided for free added substantial value for a particular customer, so it modified its approach to pricing and selling its services accordingly.

It may require time to introduce and implement value-based approaches in large contracts; the case firms described being "on a learning journey" or "experimenting with new ways" of operating. It is challenging to shift the dialogue with customers to include advanced, process-delegation services (i.e., activities that customers otherwise would perform themselves). This evidence implies an intricate relationship between identified value and value demonstrated through service delivery, such that the customer gradually comes to realize the provider's capabilities.

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Finally, learning involves developing an ability to sell additional services, beyond existing contracts, as customers gradually become more experienced with services and solutions. Such learning supports dialogue and the sale of value-based service agreements. Similarly, customers explained that after complex equipment was in operation, they grew more aware of their need to understand how they could benefit further from the provider's offerings, which expanded the scope for potential service agreements to address customers' needs, as well as creating further entry barriers to competitors.

4.4.2 Relational processes. Solutions may be predicated on customer–supplier relational processes, aimed at addressing customers' business needs at profitable prices (Tuli et al., 2007). The management of customer relations thus is an important part of VBP and VBS capability developments. Both providers used information systems to manage customers. Alpha's information system documented services to capture essential aspects of the customer interactions and create information sharing routines between service and sales engineers. Beta developed standardized routines and protocols for equipment inspections, so that it could capture and store more essential information to manage customer relationships. Moreover, Beta established a training academy for customer development, which helped it build relations and overcome customer apprehension about its increasing role in activities previously performed in-house. Alpha trained its service engineers to perform on-site development with customer staff while they also performed repair and maintenance activities in locations that were not easily accessible or were limited to accredited personnel. To this end, scripted routines developed over time, which also allowed Alpha to identify opportunities for service and gain greater access key decision makers.

Value discovery thus arises from relational processes in which providers and customers jointly develop stronger understanding of value. Providers must manage customer relationships while selling contracts, but they also need to engage with subsuppliers. Even

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despite their stated concerns about sharing value and cost information, providers, subsuppliers, and customers recognized the great potential for partnerships and joint learning. The primary research focus here is the efforts of Alpha and Beta, but the findings suggest that for VBP and VBS, it would be important to develop close relationships with subsuppliers and include them in dialogues with customers to assist with the discovery and delivery of value.

Another challenge entails managing resistance to advanced services among employees in the customer organization, some of whom may fear replacement by personnel from the service provider. For example, Beta respondents mentioned the need to "tread delicately" in customer organizations to foster trust. Failing to manage the customer firm and its employees also may have implications for gain-sharing arrangements, if customer employees sabotage the relationship or disrupt the full benefits of a service or solution. Therefore, both Alpha and Beta sought to nurture their relationships with individual customer representatives involved in service sales and delivery. They noted the limitations of relationships at lower levels in customer organizations and the need to build on them, using face-to-face meetings to establish relationships at higher levels.

Figure 2 illustrates the identified VBP and VBS capability phases and dimensions. It also depicts how higher-level learning capabilities influence the development of these capabilities in an ongoing, iterative process.

INSERT FIGURE 2 ABOUT HERE

5. Discussion and Conclusions

Previous studies indicate that VBP and VBS capabilities are important for value creation, but they tend to separate these capabilities into pricing (e.g., Rapaccini, 2015; Sharma & Iyer, 2011) and those related to selling (e.g., Kindström et al., 2015; Ulaga & Kohli, 2018; Ulaga & Loveland, 2014), or else discuss them as part of a larger research framework (e.g., Storbacka, 2011; Ulaga & Reinartz, 2011). This study adds to the available empirical insights

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by examining *both* pricing and selling capabilities and thereby identifying phases and dimensions of these capabilities and the influence of a higher-level learning capability in developing them. The results indicate that VBP and VBS are interlinked, requiring the development of various tools, skills, routines, and processes. Although most capabilities relate to both VBP and VBS, some are specific to VBS (i.e., sales tools and value proposition development). In Table 4, we summarize how the different dimensions relate to VBP and VBS.

5.1 Theoretical implications

This study makes several contributions to extant research, especially by identifying the process by which VBP and VBS capabilities develop. We uncover three distinct phases and their dimensions: customer identification and analysis (customer segmentation, customer value analysis, development of sales tools, value proposition development), implementation of VBP and VBS (fostering trust and building reputation, value creation, communication), and verification of value (quantification and value assessment, value capture, knowledge management) (Fig. 2). The findings affirm previous predictions of similar phases and capabilities (Steiner et al., 2016; Terho et al., 2012; Töytäri & Rajala, 2015), though with some notable differences. Specifically, Töytäri and Rajala (2015) propose a leverage phase, but we find that the verification of value and its distinct dimensions (quantification, value assessment) (Hinterhuber, 2004, 2017), as well as value capture, mainly occur after the implementation phase. Moreover, knowledge management emerges as an important dimension, broader than just developing a case repository (Storbacka, 2011; Töytäri & Rajala, 2015). The different capability dimensions we identify also are underpinned by a higher-level learning capability, which entails value discovery and relational processes.

Both VBP and VBS capability phases and dimensions occur iteratively and overlap; in this sense, this study validates and extends the range of capabilities identified in prior

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literature (Terho et al., 2012; Töytäri & Rajala, 2015). Whereas extant studies imply an ossification of capabilities and managerial practices within distinct phases or stages, we show that they are more fluid, such that capability dimensions across different phases become imbricated. For example, developing sales tools for a customer TCO analysis during the customer identification and analysis phase is important for VBS, in that it enables a better understanding of the potential ways offerings can contribute to value creation and communication in the implementation phase. Moreover, sales tools support the application of TCO models, in dialogue with customers at an operational level, which is important to ensure value is commensurate with customer needs during the selling process and to support value verification. These tools get refined over time, based on their use in practice, which reveals an important connection between VBP and VBS. By examining both capabilities, this study supplements the few contributions that have explored their relationship.

Another contribution is that we establish how learning influences VBP and VBS capability developments. Higher-level learning challenges organizations to consider the need for change and seek novel ways to do things (Cope, 2003; Engström & Kerosuo, 2007). Pricing and selling industrial services based on value requires a break from traditional paths, and the identification of value-in-use is an essential part of this break. For industrial services and solutions, value does not exist "out there" to be discovered; instead, they require learning processes in which the provider and customer jointly realize and co-create potential value. This conceptualization of value suggests a new view on how industrial services and solutions can improve a customer's operations: by establishing access and developing opportunities to engage in value discovery. As the cases illustrate, this type of redefinition involves developing relationships with new parts of the customer organizations and acquiring access to higher-level decision makers. Both the discovery and the delivery of value involve changing

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the modus operandi of the firm, such that VBP and VBS require learning from feedback from customers and suppliers and learning-by-doing (Gibb, 1997; Zollo & Winter, 2002).

Although the identified phases are important for identification, analysis, implementation, and verification of VBP and VBS, success is likely contingent on expansive learning (Engeström, 2014). Winter and Szulanski (2001) discuss potential replication strategies that rely on abilities to experiment, develop, and replicate the essential elements of a business model (including pricing) while also adjusting these abilities to new contexts. In this regard, learning processes support the discovery of value in relational processes, which are integrated with parts of the organization responsible for service delivery. Integration occurs not only between pricing and selling services but also among the customer, provider, and subsupplier, leading to modified value constellations (Normann & Ramirez, 1993) and mutual adjustments to the business relationships of different actors, so that their joint knowledge and competencies reveal new collaboration channels. Understanding the role of pricing and selling capabilities requires recognition that learning is essential to industrial services and solutions, through the experimentation, discovery, adaptation, and application of VBP and VBS in different contexts. Adding to key insights about the necessary capabilities for adjacent stages of VBP and VBS (e.g., Terho et al., 2012; Töytäri & Rajala, 2015), we argue that these capabilities are necessarily underpinned by learning.

Finally, in extending research that demonstrates the importance of developing VBP and VBS capabilities (e.g., Hinterhuber, 2017; Liozu & Hinterhuber, 2013a, b; Töytäri et al., 2015; Töytäri & Rajala, 2015), we highlight how the higher-level capability of learning enables value discovery with the customer, as well as the replication and adaptation of operational VBP and VBS capabilities to different contexts. As a theoretical implication, we show that industrializing advanced services can be facilitated by the development of scalable

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and replicable learning routines and a firm's ability to deploy its capabilities in other contexts, despite the contextual or customized nature of industrial services and solutions.

5.2 Managerial implications

From a managerial perspective, the findings suggest that in pursuing VBP and VBS for industrial services and solutions, firms must make significant investments to develop the capabilities needed for identifying, implementing, and verifying value. These identified dimensions are interrelated and should persist across the organization. That is, pricing and selling cannot be understood separately but rather require interlaced knowledge across marketing, sales, and service units, with a clear acknowledgement of the importance of discovering value-in-use through customer relationships.

Managers can benefit from understanding broader implications of VBP and VBS, including how their implementation is linked to other firm functions and to the customer. Value discovery does not occur solely in R&D labs or business development functions. It requires relational processes with customers that unfold over time, such that significant problems can be matched with solutions that build on the provider's existing capabilities and also meet customers' perception of value. Complexity arises when customers differ in their competencies and preferences, which inform their willingness and capability to engage in dialogues about value. A provider may wish to discuss value; the customer may be focused on price or lack appreciation for the value potential of comprehensive services and solutions. Conversely, the provider may lack detailed insights into the customer's business and needs, such that it cannot provide adequate value propositions (Terho et al., 2012). From a relational perspective, it is necessary to design strategies to manage dialogues and develop customer insights, along with coordination mechanisms, systems, and routines that support value discovery for price setting (Dutta et al., 2003). This managerial task requires significant

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relational skills and sensitivity, as well as detailed insight into the firm's own capabilities, which must be demonstrated to foster trust and build reputation with customers.

Knowledge resulting from value discovery and relational processes shapes VBP and VBS capabilities, supporting their replication and adaptation to other contexts. Therefore, VBP and VBS should be key inputs for new service development, according to the ability to realize value (Johansson, Keränen, Hinterhuber, Liozu, & Andersson, 2015; Liozu, 2015b). In contrast with new products that can be tested, standardized, and quality-assured in advance, a service is not "fixed" or ready, even when it reaches the sales stage. Instead, local adaptations often are needed to address customer- or segment-specific needs. The first version of a service also may resemble a prototype, without a clear pricing model or sales approach; they only become clear and more formalized as the service gets increasingly sold, delivered, and used in the field, allowing for the realization of value. Nor is value static; customers' perceptions change over time (Flint et al., 2002). The discovery of value and learning thus are both critical. For industrial firms, successful engagement in VBP, VBS, and learning can redefine markets, both by changing the target decision makers within the customer organization and through recombinations of services to form novel solutions to customer-specific problems. However, customers do not simply agree to engage in dialogue about VBP; a value-based approach demands substantial effort from the provider. To extend the dialogue to include higher-value offerings, it must offer evidence of the firm's ability to deliver value by solving customers' problems. Sharing knowledge and learning from past experiences, on ongoing bases, thus are important techniques, whereas reliance on ad hoc problem solving is insufficient.

5.3 Limitations and further research

This study is not without limitations. It is based on an in-depth study of two organizations, offering only a limited degree of analytical generalizability to other industrial firms. Further

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studies might explore VBP and VBS capabilities for a broader range of industries and contexts, to determine the extent to which the findings we present apply to organizations in other industries. We also call for studies of the role of customers and suppliers and their implications for VBP and VBS. As Kienzler and Kowalkowski (2017) suggest, studies that adopt a buyer perspective could offer unique insights into customers' perceptions of value. Such approaches might complement the predominant provider perspective in extant literature and enhance understanding of the need for industrial firms to develop capabilities to create value in partnership with external actors. Researchers also might investigate the learning processes organizations adopt to implement VBP and VBS. Such studies would likely require other approaches to understanding pricing decision making and perceptions of value, such as the repertory grid technique (Goffin et al., 2012). By building on this study, we hope continued research enhances theoretical and managerial understanding of value-based approaches even further.

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Appendix 1. Sample interview guide

Organizational and personal history

- Can you tell us about your background and how you came to work in your current role?
- Can you describe the organizational structure? Has it changed in recent times?

Strategic overview

- How would you describe the development of the global market in your industry over the last 10 years?
- What challenges have the company experienced in adapting to new markets?
- What are the strategic priorities of your company?
- How would you describe your business model(s)?
- Can you describe the markets in which you operate? What changes have you seen over time?

Provision of industrial services and solutions

- What products does your company provide? Please describe the different contexts in which your products are used.
- What services does your company provide? How do you categorize services?
- How do you define customer solutions?
- How do you integrate products and services to provide customer solutions? What new capabilities do you require to provide solutions?

Pricing and selling – products, services, and solutions

- In your position, how do you engage in pricing and selling?
- How do you price and sell your products?
- How do you price services? How do you price solutions?
    - Please tell us about the last time you were involved in a price decision.
    - Do prices differ, for example, among customers? Why?
- Please describe the sales process for a service and solution from start to end based on a recent example.
- What factors do you think are important for making a price decision?
- What challenges do you encounter when pricing and selling offerings?
- What do you do when you encounter uncertainty or doubt?
- What kind of justification is frequently used in the pricing and selling process?
- Please elaborate on the role of value in pricing and selling.
    - How do you define VBP and VBS?
- To what extent have you adopted and implemented VBP and VBS?
    - What are the risks and difficulties of doing so?
    - What role does the customer play in VBP and VBS?
- How do you assess and demonstrate value to your customers?
- What pricing tools/systems/rules/guidelines are in place?
- Please elaborate on your practices for monitoring and controlling prices and sales numbers.

Others

Is there anything we have overlooked? Is there anything you would like to add?

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Appendix 2. Coding categories and illustrative quotes

Second-order themes	*Illustrative quotes*
Customer segmentation	• "The dimensions that we are segmenting on are recurrent based business with the client and also the size of the client, especially the size of the business. We exclude some areas, and mainly for political reasons, if we are uncertain that we can get the money and if we are not sure…that our employees can stay safe in the area." (Vice President, Alpha) • "…the big difference between wind farms and the oil industry is that the wind industry does not have [its] own technicians who can travel out and handle the systems. But our customers in the oil industry, they have their own who can handle the equipment, and that's a big difference." (Service Manager, Alpha)
Customer value analysis	• "… if you spend $½ million… and… they [the customers] … go, 'Where's the value in that? It's not worth $½ million'. And we say, 'Well, by providing it, we'll be able to reduce your operating costs by half to $1 million a year', and so they see the value. So, we would value price it, rather than just look at what it costs us and add a margin to it." (General Manager, Beta) • "…we are able to map complete plants to conduct value analysis" (Director of Services, Beta) • "…we are doing 3D-scanning of offshore installations to see where we can add value (Business Development Manager, Alpha)
Development of sales tools	• "I look at the TCO tool and…what we can do now is that at least we have better forecasting capability in terms of […] making sure we have the spares that we need when we need them." (General Manager, Beta) • "We have developed sales tools that we are training the salesforce to use…they can then tailor these to specific plants to show savings" (General Manager, Beta) • "we are able to do models in the simulations we do to show the value" (Business Development Manager, Alpha)
Value proposition development	• "I typically do this drawing when I'm up against local competitors [pointing at drawing on the white board]… A bottleneck over here, I think you have potential issues in your roll mill because the way you burn your clinker indicates you're running the roll mill wrongly. … this is about linking our full flow sheet capabilities to some of our single services simply to make a value proposition that [the] local competitor…cannot talk." (Department Head, Beta) • "The service products should be designed into the solutions that we sell. Right. But that takes the core company to understand the concept of selling service as part of our complete business. Right now, they are very project focused." (Sales Manager, Workshop, Alpha)
Fostering trust and building reputation	• "They've got to trust you, so you've got to get close to the customer. They have to understand who your organization is, where you've done things before. So, it's a different type of selling model.…" (General Manager, Beta) • "…They [service engineers] are talking to the customer, and the customer trusts them because they are out on a visit, they are always solving a problem and it's a good way trusting people." (Service Manager, Alpha) • "Sometimes I'm allowed to talk about it. It depends on the mood [of] the plant manager, the people I meet on site and so it's very much about getting a sense of where are we on this medium. …That's about having an eye for that. If I can have this dialogue with a customer then they're not even going to think about the small local competitors because then they perceive us as someone who is able to add more than just value on a single service." (Department Head, Beta)
Value creation and communication	• "I think you need to create value for the customer. I think you need to be able show what value you're creating, I think that's the first thing you have to do. If you get that right, then it doesn't really matter what you do." (Innovation Manager, R&D, Beta) • "…when you have no description of what we're getting and we have no knowledge of what value it is this brings and…then you'll never get somebody paying for it." (Manager, Customer 3) • "…actually get the customer involved in the test work, get them to understand what the process is that we're supplying them, give them examples of where its worked before, so references… if you're providing something on a value-base, the customer has to have a comfort level that there is no risk around it." (General Manager, Beta) • "…it is complicated to explain to the customers what they really get before they are paying." (Bid and Proposal Manager, Alpha)
Quantification and value assessment	• "That's also extremely important so we can put the documentation into our maintenance system, saying on a later stage, if we have an issue with the area we have been working in, then we can pull out a report and show to our clients, 'Well, it was by a qualified welder, and here's his welding certificate, and all was good.'" (Procurement Manager, Customer 1) • "…you can look at the entire budget for our system and say, well how much are you willing to pay per year for this service? But how much? …I think it should be based on the, yeah, what time does it require. And

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	then set a price for that because, yeah, there's two ways you can sell things. You can sell it based on the cost. And you can say, sell it based on the value." (Manager, Customer 3) • "These KPIs change quite rapidly. Right now, we have a contract in Norway. We had the discussions with them; everything was good and they were selling quite well. Now the oil prices has just gone to the bottom . . . now they had one stop, which was longer than we thought." (Innovation Manager, R&D, Beta) • "Because if we choose to buy a radio communication system, which is, as Tetra is, extremely expensive…well, are we willing to pay for the service that's required for this? Yes, of course, because it's our last line of communication for these guys we have out there. And, communication is part of their security. So, yes, it is important to us that we do this". (Customer 2, Communication Engineer)
Value capture	• "[T]hat's really one of the biggest problems in the Service Department, that we are helping them for free, when they are phoning in." (Service Manager, Alpha) • "…we model what we expect to make from the different service offerings" (Sales Manager, Beta) • "…as a sales person you need to be good at negotiating a fair amount for the value created" (Sales Manager, Alpha)
Knowledge management	• "...usually the site manager is a mechanical engineer of some kind. He has no knowledge whatsoever about these kind of systems, so usually it's really hard to persuade them to do a service and maintenance contract because they have no idea what they're getting or what they are buying. And, this has been also one of the learnings I think that you can get from [Alpha], it has been quite hard to get [us] to sign these service contracts." (Communication Engineer, Customer 2) • "...our customer relationship management tool provided by Microsoft, now has service management tool that's part of it. So we think that we're going to be getting that very soon. We're looking forward to that and hoping that that's going to help us to manage our global service resources a lot better. … Scheduling, providing location to track skills, development, providing a location for a knowledge center to keep all kinds of documents, drawings and instructions manuals, all of that at easy reach for the field engineers." (Vice President, Beta) • "So all the knowledge on the market is inside of the head of people. . . . Of course we are still people and we still talk but it is a major challenge for the market side that we do not have a complete overview. And if we lay off people, we lay off knowledge." (Vice President, Workshop, Alpha)
Value discovery	• "Then we calculated the value for these, and it was something like two hundred million DKK [33 USD million] a year in savings, and we did it for free." (Innovation Manager, R&D, Beta) • "We are dealing with customers that have a long history and tradition in doing all the things that we want to do, which they of course, and with some right, think that they're doing better than we can do. So, that's a question about the building of confidence in the market, and that's a heavy task I would say, but we can do it . . . if you have Field Service Engineers, assistance through contracts over years, for key machinery, and . . . turnkey services." (Department Head, Beta) • "We have actually sold additional services for approximately 30% to 40% of the contract value on top of the service agreement. Maybe that's at least an internal learning. . . . We've actually created an additional sales and at the same time, made sure that we build up entry-barriers for some of our competitors because they have a service agreement with [Beta]." (General Manager, Beta)"
Relational processes	• "We have a maintenance superintendent who didn't want us on site. He saw [us] stepping into his area to prove that he was unqualified to do his job." (General Manager, Beta) • "I want you to look good in front of your customer. So whatever we can do to help support you on site meetings—coming without any [supplier] badges or anything, just come along to see it and answer technical questions or do a short presentation—we're willing to do that." (Sales & Marketing Manager, Supplier 2) • "To try to do an impact, you've got to be part of the team at site." (Innovation Manager, R&D Beta) • "We need to go to the site. We need to go to talk with the superintendents, the managers, [to] get information about what they are thinking." (Service Manager, Beta)

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Table 1: Overview of case firms

Firm		Core business/sector	Typical services	Number of Employees (Approximate)	Turnover (Approximate)	Percentage Turnover from Services
Alpha		Systems integration, manufacture of firefighting and telecommunication equipment	Support and service contracts, service hotline, installation and commission, repair and maintenance, system monitoring, remote support, preventative maintenance, training services, e-learning, maintenance strategy development	1,200	225 million USD	20% (45 million USD)
Suppliers	Supplier 1	Firefighting systems	Repair and maintenance, spare parts, technical support, periodic service, service agreements	60	Gross profit 9 million USD	Not available
	Supplier 2	Firefighting systems	Training, technical support	400	100 million USD	Not available
	Supplier 3	Telecommunication equipment	Not applicable	50	Gross profit 9 million USD	Not applicable
Customers	Customer 1	Energy utility	Not applicable	5,700	10 billion USD	Not applicable
	Customer 2	Drilling operator	Not applicable	Not available	2 billion USD	Not applicable
	Customer 3	Energy utility	Not applicable	20,000	17 billion USD	Not applicable
Beta		Cement and mining plants	Operations and maintenance contracts, shutdown services, spare parts delivery, training institute, remote monitoring, preventative and predicative maintenance	12,000	2.7 billion USD	56% (1.5 billion USD)
Customer	Customer 4	Mining customer	Not applicable	5,600	6 billion USD	Not applicable

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Table 2: Data sources

*Actual duration of audio recordings.
†Audio was not recorded, but detailed notes were taken.

Case Firm	Data Source	Position of Informant	Number of Interviews/ Workshops	Duration (mins)*
Alpha	Interviews	Vice presidents	5	[68, 47, 61, 41 + 77]
	Interviews	Sales managers (area manager, after-sales)	3	[59, 63, 63]
	Interviews	Service manager and coordinators	4	[88, 49, 88, 42]
	Interviews	Project managers	2	[71, 50]
	Interviews	Service engineers	3	[49, 61, 46]
	Interviews	Managers (senior, bid & proposal, procurement, product responsible, business development, commissioning, tender managers)	8	[56, 57, 69, 53, 227, 54, 52, 76]
	Interview	Logistics assistant	1	[51]
	Group interview	Department head, service managers	1	[175]
	Workshop	Vice presidents, service managers, sales managers	2	[240†, 120]
	Supplier 1 – Interviews	Sales manager, global supply chain manager, commercial manager	3	[57, 47, 37]
	Supplier 2 – Interviews	Sales & marketing manager, service manager	2	[67, 67]
	Supplier 3 – Interviews	Key account manager, sales manager	2	[57, 60]
	Customer 1 – Interviews	Procurement manager, senior manager	2	[54, 60]
	Customer 2 – Interview	Communication engineer	1	[50]
	Customer 3 – Interview	Manager	1	[94]
Beta	Interviews	Vice presidents	7	[57, 101, 61, 58, 57, 72, 61]
	Interviews	Director of services	3	[60, 71, 96]
	Interviews	Department heads	4	[56 + 52, 71, 83]
	Interviews	General managers	8	[58, 57, 64, 55, 77, 66, 69, 69]
	Interviews	R&D function (technical, innovation, project managers, engineers)	9	[115, 58, 71, 48, 36, 47, 32, 39, 65]
	Interviews	Sales managers	5	[76, 64, 79, 101, 93]
	Interviews	Service managers	4	[67, 76, 59, 70]
	Group interviews	Innovation, technical managers	3	[60, 67, 60]
	Workshops	Vice presidents, managers	2	[116, 180†]
	Customer 4 – Interview	Chief procurement officer	1	[30]

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Table 3: Criteria for assessing reliability, trustworthiness, and validity of research

Criteria	Method	References
Credibility Degree to which the findings are an adequate representation of the data	• Data triangulation (interviews, documents, observations, field notes). • Iterative research process between literature and emergent findings. • Case reports provided to Alpha and Beta for comments and discussion. • Regular feedback workshops conducted with Alpha and Beta. • Ongoing meetings with representatives throughout the research. • Feedback from case company employees at academic conferences.	Lincoln & Guba (1985); Miles & Huberman (1994)
Transferability Extent to which findings can be applied to another context	• Analytical generalizability of findings to other company contexts, achieved through the presentation and discussion of findings with other Danish firms in an industry core group established as part of a wider research project.	Lincoln & Guba (1985); McCracken (1988); Goffin, Raja, Claes, Szwejczewski, & Martinez (2012); Eisenhardt & Graebner (2007)
Dependability Extent to which the findings are reliable and consistent	• Data collection log maintained throughout the research process. • Detailed record of all data sources maintained to allow for tracking. • Multiple rounds of coding by a team of researchers. • Documentation and recording of feedback provided in workshops.	Lincoln & Guba (1985); Halldorsson & Aastrup (2003)
Confirmability The interpretation of data in a logical manner, demonstrating a clear link between data and findings	• Interviews conducted at different levels, locations, functions, and times in Alpha and Beta. • Multiple researchers used to reduce researcher bias. • Findings discussed with interviewees and senior management of Alpha and Beta. • Findings audited and discussed by experts and other researchers.	Lincoln & Guba (1985); Miles & Huberman (1994); Valle, King and Halling (1989); Riege (2003)
Integrity and generality Extent to which interpretations are not biased by interviewees through inflated accounts, misinformation or withholding of information. The use of multiple perspectives to understand phenomenon being researched	• Interviewees provided briefing notes prior to interviews to allay any concerns and provide background for project. • Rapport built with interviewees. • Interviewees guaranteed anonymity. • Multiple rounds of interviews to check for inconsistencies in explanations. • Researchers immersed to varying degrees in Alpha and Beta. • Use of other data sources, including observations of meetings (departmental, sales, project-specific) and documentation (including pricing information, marketing material). • Interviews conducted with customers and suppliers of Alpha to corroborate (triangulate) accounts. • One of the researchers has work experience as a value and pricing analyst and pricing consultant and is a regular speaker at pricing manager conferences and thus provided external, practical know-how.	Wallendorf & Belk (1989); Strauss & Corbin (1990); Flint. Woodruff, & Gardial (2002)
Fit Extent to which the findings fit within the area of study	• Careful selection of cases to ensure comprehensive picture of VBP and VBS in the context of industrial offerings. • In-depth interviews with specific individuals working with VBP and VBS in Alpha and Beta (e.g., salespeople, marketing personnel, pricing and sales tool developers, business development team).	Strauss & Corbin (1990); Flint et al. (2002)
Understanding Extent to which findings are accurate representations of the reality experienced by participants	• Findings presented and discussed at Alpha and Beta throughout research process. • Findings presented at two academic conferences. • Findings found to be relevant by other companies at industry meetings and academic conferences.	Strauss & Corbin (1990); Flint et al. (2002)

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Table 4: Identified capabilities for VBP and VBS

	Value-Based Pricing	Value-Based Selling
Customer identification and analysis		
Customer segmentation	*Offering distinctions:* Insights from customer segmentation are needed to create segment-specific value propositions and pricing structures that link customer segments to specific service offerings.	*Knowing whom to target:* Customers' technological competency and attitude toward maintenance and service activities reveal specific target segments for value-based offerings and influential stakeholder groups within target firms.
Customer value analysis	*Knowing value logic:* Understanding customer processes and operations is a means of knowing where value can be provided and charged for, rather than pricing based on costs and mark-ups.	*Spotting opportunities:* Sales personnel gain more insights from sales activities, because they have specific, detailed understanding of customers' production processes, allowing them to identify opportunities for improvement.
Development of sales tools		*Value tools:* Sales personnel should be armed with tools and devices, such as total cost of ownership analyses, animated movies, 3D scans, and models, to illustrate potential for value creation through services.
Value proposition development		*Timing of sale:* Services and solutions need to be part of the sale of products and projects (e.g., in the tendering process) with a clear outline of benefits.
Implementation of VBP and VBS		
Fostering trust and building reputation	*Perceived fairness:* Awareness of pricing fairness and the impact on customers' perceptions of the trustworthiness and reputation of the firm.	*Creating dialogue:* Develop a close dialogue with key decision makers over time by demonstrating know-how and the ability to execute projects in ways that enhance customer value.
Value creation and communication	*Build value metrics:* Develop metrics, related information systems, and routines for capturing performance enhancements to enable value creation as well as the communication of value to customers.	*Communicating value:* Ability to communicate the (potential) value created for the customer to sell and/or renew contracts.
Verification of value		
Quantification and value assessment	*Analytical process:* Processes to evaluate value delivered to the customer through service contracts and the provider's profit. Such assessments may have an impact on the KPIs and be subject to change.	*Showcase value:* Important analytical means for making delivered benefits visible to customers. Providers must quantify the results of service provision to enable value assessment.
Value capture	*Valuate contract:* Ability to receive a price commensurate with the value delivered.	*(Re)negotiate sharing:* Ability to (re-)negotiate and agree on service contracts to ensure that the price matches the customer's willingness to pay and reflects a fair share of the value created.
Knowledge management	*Refine pricing:* Ability to share and facilitate knowledge in relation to pricing aspects is key for improving a (new) offering (e.g., KPIs for gain sharing purposes).	*Capture know-how:* Ability to identify new business opportunities and create and extend sales of projects, products, and services. VBS opportunities can be identified through experiences of personnel involved in delivery of a service.

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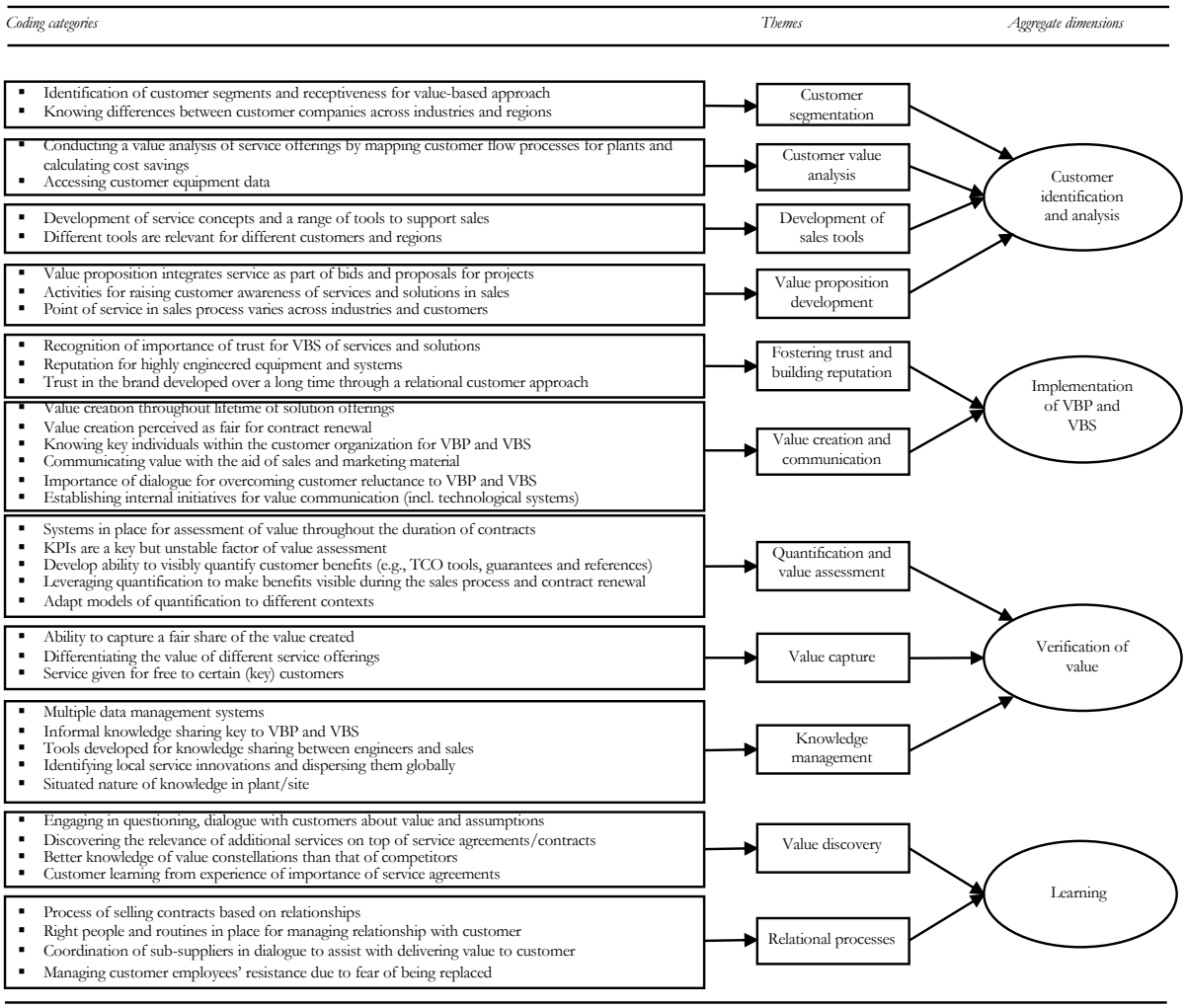

Figure 1: Data structure

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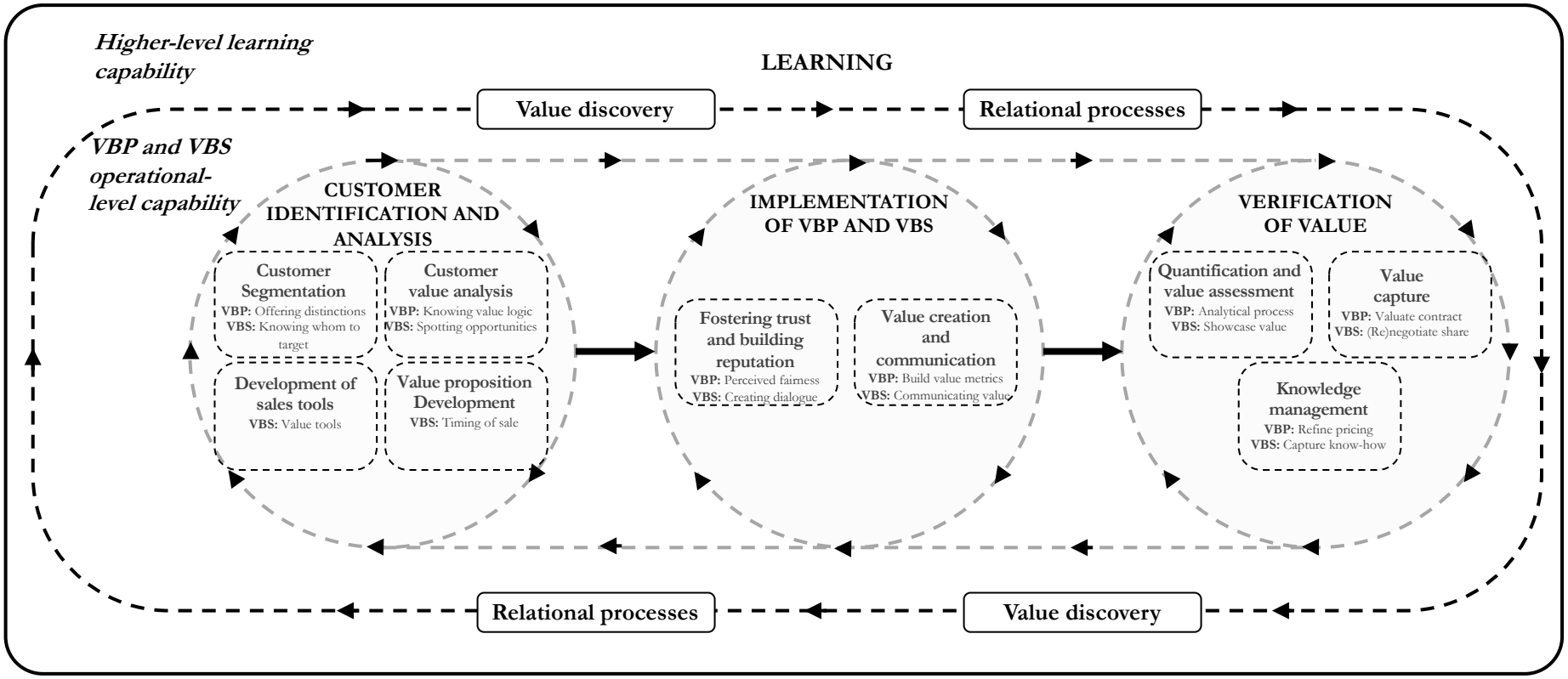

Fig. 2. VBP and VBS capability development and the influence of learning

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